A Cross-AP Position-Fused Channel Estimation Framework for Non-Uniform Sparse Arrays in Near-Field Cell-Free XL-MIMO

Step-by-step mathematical model (paper-ready, simple wording)

Mathematical framework note

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How to read this. Every block gives (i) the equation, (ii) one line of plain-English meaning, and (iii) the reference. Equations tagged **[new]** are the contribution of this work (the *CAPE* estimator); all others are standard and are cited. The notation matches the simulation code one-to-one, so the figures can be reproduced from these formulas.

1 Notation

L	number of APs	K	number of users
N	antennas per AP (sparse ECA)	$\lambda,\ d_0 = \lambda/2$	wavelength, half-wavelength
$\mathbf{x} = [x_1, \ldots, x_N]^{\mathsf{T}}$	antenna positions on the array [m]	$D = x_N - x_1$	array aperture [m]
$\mathbf{c}_l \in \mathbb{C}$	horizontal location of AP l	$\mathbf{u}_k \in \mathbb{C}$	horizontal location of user k
Δh	AP–user height difference	$r_{l,k},\ \theta_{l,k}$	3-D distance, elevation
$\varphi_{l,k}$	azimuth of user k at AP l	$s_{l,k} = \sin\varphi_{l,k}\cos\theta_{l,k}$	effective sine
$\beta_{l,k}$	large-scale gain (incl. noise norm.)	p	normalized pilot/data SNR
$\mathbf{h}_{l,k} \in \mathbb{C}^N$	channel AP $l \to$ user k	$\sigma^2 = 1/(p\tau)$	post-despreading noise power
$d_R = 2D^2/\lambda$	Rayleigh distance	$\mathcal{A}_{l,k}$	near-field indicator ($r_{l,k} < d_R$)

2 System model

2.1 Cell-free layout

L access points (APs), each with a sparse array of N antennas, jointly serve K single-antenna users and are connected to a central processing unit (CPU) [1, 2]. We study the uplink. The APs and users are dropped in a square area; $\mathbf{c}_l$ and $\mathbf{u}_k$ denote their horizontal locations.

2.2 Non-uniform sparse array (ECA) geometry

Each AP uses an *extended co-prime array* (ECA) [3], a non-uniform sparse array built from the co-prime idea [4]. Take two integers $M, \tilde{N}$ with $\gcd(M, \tilde{N}) = 1$. One "block" is the union of two uniform rulers sharing their first element,

$$\mathcal{B} = \{\, m\tilde{N} :\ m = 0, \ldots, M-1 \,\} \cup \{\, nM :\ n = 0, \ldots, \tilde{N}-1 \,\} \quad \text{(units of } d_0\text{)}. \tag{1}$$

With B blocks (even), the ECA index set is the block repeated with period $M\tilde{N}d_0$ and mirrored about the origin,

$$\mathcal{N} = \bigcup_{b=0}^{B/2-1} \Big\{ \pm \big(b\,M\tilde{N} + a\big) : a \in \mathcal{B} \Big\}, \qquad x_n = \big(\mathcal{N}_n - \bar{\mathcal{N}}\big) d_0, \tag{2}$$

where $\bar{\mathcal{N}}$ centres the array. This yields $N = B(M + \tilde{N} - 1) - 1$ elements and aperture $D = (BM - 2)\tilde{N}d_0$ [3]. *Plain words:* the antennas sit at *unequal* gaps, so the array is wide (large D) but has no exact periodicity. *Example (simulation):* $M = 5, \tilde{N} = 4, B = 8 \Rightarrow N = 63$, $D = 7.6\,\mathrm{m}$.

2.3 Near-field (spherical-wave) channel

A user in the near field ($r_{l,k} < d_R$) is seen under a *curved* wavefront, so the per-antenna distance varies across the array. For AP l, user k, antenna n [5, 8],

$$r_{l,k,n} = \sqrt{r_{l,k}^2 + x_n^2 - 2\, r_{l,k}\, x_n\, s_{l,k}}\,, \tag{3}$$

and the (line-of-sight, non-uniform spherical wave, NUSW) channel is

$$\boxed{\mathbf{h}_{l,k} = \sqrt{\beta_{l,k}}\; \mathbf{a}\big(r_{l,k}, s_{l,k}\big), \qquad \big[\mathbf{a}(r,s)\big]_n = \frac{1}{\eta}\,\frac{r}{r_n}\, e^{-\jmath\frac{2\pi}{\lambda}(r_n - r)},} \tag{4}$$

where r_n follows (3) and η normalizes $\|\mathbf{a}\| = 1$. *Plain words:* the channel is fully described by the user's *position* through the pair $(r_{l,k}, s_{l,k})$ — a key fact CAPE exploits. The amplitude term r/r_n is the near-field (spherical) correction; setting $r \to \infty$ recovers the plane-wave (far-field) steering vector [5].

2.4 Far-field links and hybrid classification

A link with $r_{l,k} \geq d_R$ is far-field and modelled as correlated Rayleigh fading, $\mathbf{h}_{l,k} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_{l,k})$, with a local-scattering covariance built from the element positions [7]. The indicator $\mathcal{A}_{l,k} = \mathbb{1}\{r_{l,k} < d_R\}$ is a large-scale quantity known at the CPU. Because $d_R = 2D^2/\lambda$ grows with the *square* of the aperture, a wide sparse array makes far more links near-field (hybrid regime) [3, 8].

3 Pilot phase and the sparse-array ambiguity problem

3.1 Received pilots

With mutually orthogonal pilots of length $\tau = K$ [1], after correlating with user k's pilot, AP l obtains the despread observation

$$\mathbf{z}_{l,k} = \mathbf{h}_{l,k} + \mathbf{w}_{l,k}, \qquad \mathbf{w}_{l,k} \sim \mathcal{CN}\big(\mathbf{0}, \sigma^2 \mathbf{I}_N\big), \quad \sigma^2 = \frac{1}{p\,\tau}. \tag{5}$$

Plain words: each AP gets a noisy copy of the channel it must estimate.

3.2 Polar-domain representation and why a sparse array is ambiguous

Collect candidate atoms $\mathbf{a}(r, s)$ on a grid of angle $s \in [-1, 1)$ and inverse range $q = 1/r$ into a polar dictionary, the standard near-field sparsifying basis [5, 6]. The *spatial response* between a candidate (r, s) and the true $(r_{l,k}, s_{l,k})$ is

$$g_l(r, s) = \big|\mathbf{a}(r, s)^{\mathsf{H}}\, \mathbf{a}(r_{l,k}, s_{l,k})\big| \in [0, 1]. \tag{6}$$

For a *dense* array g_l has a single sharp peak. For a *sparse* array g_l also has *side peaks* (ambiguity). For a *uniform* sparse array of spacing $\eta\lambda/2$ these are exact grating lobes at

$$\sin\theta_u = \sin\theta_0 + \tfrac{2u}{\eta}, \qquad u \in \mathbb{Z}\setminus\{0\}, \tag{7}$$

refocused in range at $r_u = r_0\cos^2\theta_u/\cos^2\theta_0$ [3, Lemmas 5–6]. For the *non-uniform* ECA the irregular gaps prevent the side peaks from aligning into a full-height grating lobe, leaving only *weak, scattered* side peaks [3, Sec. III].

The core difficulty. From one AP's noisy $\mathbf{z}_{l,k}$, the true peak and a side peak of (6) can look alike, so a single-AP estimator may lock onto the wrong one. This is the weakness CAPE removes.

4 Baseline: per-AP local estimation (polar-domain OMP)

The conventional estimator runs orthogonal matching pursuit (OMP) [5, 11] on $\mathbf{z}_{l,k}$, *independently at each AP*. With residual $\mathbf{r}$ (init. $\mathbf{r} = \mathbf{z}_{l,k}$):

$$(\hat{s},\hat{q}) = \arg\max_{(s,q)} \left|\mathbf{a}(s,q)^{\mathsf{H}}\mathbf{r}\right| \quad \text{(peak + gridless refinement [5])}, \tag{8}$$

$$\text{accept iff} \quad \left|\mathbf{a}(\hat{s},\hat{q})^{\mathsf{H}}\mathbf{r}\right|^2 \;\geq\; \gamma\,\sigma^2, \qquad \gamma = \ln D_{\text{dic}} + 0.58 + c, \tag{9}$$

where (9) rejects noise-only peaks: $\gamma\sigma^2$ is the level the largest of D_{dic} independent noise correlations reaches (extreme value of D_{dic} exponentials; $c \geq 0$ a margin) [11]. Selected atoms form $\mathbf{G}$; the channel and its error covariance use the data-estimated path powers (LMMSE/Wiener weighting),

$$\hat{\mathbf{h}}^{\text{loc}}_{l,k} = \mathbf{G}\big(\mathbf{G}^{\mathsf{H}}\mathbf{G} + \sigma^2\text{diag}(1/\hat{\mathbf{p}})\big)^{-1}\mathbf{G}^{\mathsf{H}}\mathbf{z}_{l,k}, \qquad \mathbf{C}_{l,k} = \varepsilon^2_{l,k}\,\mathbf{I}_N. \tag{10}$$

Limitation on sparse arrays: on a weak link, (8) may select a *side peak* of (6); the resulting $\hat{\mathbf{h}}^{\text{loc}}$ points the wrong way, which is why the plain ECA has a higher NMSE than the dense array (observed in the results).

5 Proposed: Cross-AP Position-Fused Estimation (CAPE)

Idea. A user has one physical location $\mathbf{u}_k$, and *every* AP's near-field channel (4) is a deterministic function of it. Rather than let each AP guess alone, the CPU finds the *single* location on which the strongest APs *agree*; a side peak of one AP does not agree with the others, so it is rejected. The agreed location then regenerates each AP's channel.

5.1 Position parameterization

For any candidate horizontal location $\mathbf{u} \in \mathbb{C}$, AP l's geometry is deterministic:

$$d_l(\mathbf{u}) = |\mathbf{u}-\mathbf{c}_l|, \quad r_l(\mathbf{u}) = \sqrt{\Delta h^2 + d_l^2(\mathbf{u})}, \quad s_l(\mathbf{u}) = \sin\big(\angle(\mathbf{u}-\mathbf{c}_l)\big)\,\tfrac{d_l(\mathbf{u})}{r_l(\mathbf{u})}, \tag{11}$$

and the corresponding unit atom is $\mathbf{a}_l(\mathbf{u}) \triangleq \mathbf{a}\big(r_l(\mathbf{u}), s_l(\mathbf{u})\big)$. *Plain words:* pick a point on the map, and every AP knows exactly which beam would point there.

5.2 Step 1 — candidate locations from local peaks [new]

Let $\mathcal{C}_k$ be the J strongest local peaks (8) of the N_c strongest near-field APs of user k. A polar peak $(\hat{s},\hat{q})$ at AP l maps to *two* mirror locations (a linear array cannot tell front from back):

$$\hat{r} = 1/\hat{q}, \quad d = \sqrt{\hat{r}^2 - \Delta h^2}, \quad \hat{\varphi} \in \Big\{\arcsin\tfrac{\hat{s}\,\hat{r}}{d},\ \pi - \arcsin\tfrac{\hat{s}\,\hat{r}}{d}\Big\}, \quad \mathbf{u} = \mathbf{c}_l + d\,e^{\jmath\hat{\varphi}}. \tag{12}$$

These form the candidate set $\mathcal{X}_k$ (true location + a few false ones).

5.3 Step 2 — cross-AP fused likelihood [new]

Let $\mathcal{F}_k$ be the N_{AP} strongest near-field APs of user k. Score each candidate by the *non-coherent* match across those APs (each AP keeps an unknown complex gain):

$$\boxed{J_k(\mathbf{u}) = \sum_{l \in \mathcal{F}_k} \left| \mathbf{a}_l(\mathbf{u})^{\mathsf{H}} \mathbf{z}_{l,k} \right|^2, \qquad \hat{\mathbf{u}}_k = \arg\max_{\mathbf{u} \in \mathcal{X}_k} J_k(\mathbf{u}).} \tag{13}$$

5.4 Step 3 — position refinement [new]

Refine $\hat{\mathbf{u}}_k$ by a local 2-D pattern search on J_k (step δ halved on failure):

$$\hat{\mathbf{u}}_k \leftarrow \arg\max_{\mathbf{u} \in \{\hat{\mathbf{u}}_k + \delta e^{\jmath 2\pi i/8}\}_{i=0}^{7} \cup \{\hat{\mathbf{u}}_k\}} J_k(\mathbf{u}), \qquad \delta \leftarrow \delta/2. \tag{14}$$

5.5 Step 4 — per-AP channel reconstruction with a consistency gate [new]

For every near-field AP l of user k, build the atom from the agreed location, refine it in a small window ($\mathbf{b}_l^{\mathrm{fus}}$), and keep the AP's own local atom $\mathbf{b}_l^{\mathrm{loc}}$ only when it is clearly better:

$$\mathbf{b}_l = \begin{cases} \mathbf{b}_l^{\mathrm{loc}}, & \left|\mathbf{b}_l^{\mathrm{loc}\,\mathsf{H}} \mathbf{b}_l^{\mathrm{fus}}\right| \geq \mu_{\mathrm{c}} \text{ and } m_l^{\mathrm{loc}} > m_l^{\mathrm{fus}}, \ \text{ or } \ (m_l^{\mathrm{loc}})^2 - (m_l^{\mathrm{fus}})^2 > \gamma_0 \sigma^2, \\ \mathbf{b}_l^{\mathrm{fus}}, & \text{otherwise}, \end{cases} \tag{15}$$

where $m_l^{(\cdot)} = |\mathbf{b}_l^{(\cdot)\mathsf{H}} \mathbf{z}_{l,k}|$ and $\gamma_0 = \ln D_{\mathrm{dic}} + 0.58$. The channel uses a prior-based Wiener gain, and the error variance comes from the residual:

$$\hat{\mathbf{h}}_{l,k}^{\mathrm{cape}} = \mathbf{b}_l \frac{N\beta_{l,k}}{N\beta_{l,k} + \sigma^2} \left(\mathbf{b}_l^{\mathsf{H}} \mathbf{z}_{l,k}\right), \qquad \varepsilon_{l,k}^2 = \frac{1}{N} \left(\tfrac{N\beta_{l,k}\sigma^2}{N\beta_{l,k}+\sigma^2} + \left\| \mathbf{z}_{l,k} - \mathbf{b}_l \mathbf{b}_l^{\mathsf{H}} \mathbf{z}_{l,k} \right\|_{\perp}^2 \right). \tag{16}$$

Plain words: all APs rebuild their channel from the one agreed position; the gate guarantees CAPE is never worse than local estimation when the local fit is clearly stronger (high SNR). The far-field links keep the statistical LMMSE estimator [7].

5.6 Why cross-AP fusion resolves the ambiguity [new]

Take the true location $\mathbf{u}_k$ and a false candidate $\mathbf{u}'$ that is a side peak of AP $l^\star$ with sidelobe level $\rho = g_{l^\star}(\mathbf{u}') \in (0,1)$ (cf. (6)). Using $\|\mathbf{h}_{l,k}\|^2 \approx N\beta_{l,k}$ and $\mathbf{a}_l(\mathbf{u}_k) \approx \mathbf{h}_{l,k} / \|\mathbf{h}_{l,k}\|$,

$$\mathbb{E}\big[J_k(\mathbf{u}_k)\big] \approx \sum_{l \in \mathcal{F}_k} \left(N\beta_{l,k} + \sigma^2\right), \tag{17}$$

$$\mathbb{E}\big[J_k(\mathbf{u}')\big] \approx \underbrace{N\beta_{l^\star,k}\,\rho^2}_{\text{only AP } l^\star} + \sum_{m \in \mathcal{F}_k \setminus \{l^\star\}} \left(N\beta_{m,k}\,\rho_m^2 + \sigma^2\right), \qquad \rho_m \ll 1. \tag{18}$$

The false location lines up with *one* AP (ρ) but not with the others ($\rho_m \ll 1$, because a different AP sees the point at a different (r,s)), so

$$\frac{\mathbb{E}[J_k(\mathbf{u}_k)]}{\mathbb{E}[J_k(\mathbf{u}')]} \gtrsim \frac{\sum_{l \in \mathcal{F}_k} N\beta_{l,k}}{N\beta_{l^\star,k}\rho^2} \xrightarrow[|\mathcal{F}_k| \text{ large}]{} \gg 1. \tag{19}$$

Plain words: the true position wins by a factor that grows with the number of fused APs, which is exactly why a *distributed* (cell-free) system can break an ambiguity that one AP cannot. Condition (19) needs $\rho_m \ll 1$, i.e. the side peaks of different APs must *not* coincide — true for the non-uniform ECA (irregular, geometry-dependent side peaks) but weak for a uniform sparse array whose grating lobes (7) are strong and structured.

Algorithm 1: CAPE (per user k).

1. Local peaks (8) at the N_{c} strongest NF APs $\rightarrow$ candidates $\mathcal{X}_k$ via (12).
2. Fused location: $\hat{\mathbf{u}}_k$ from (13), refined by (14).
3. For each NF AP l: atom $\mathbf{b}_l$ from (15), channel $\hat{\mathbf{h}}_{l,k}^{\mathrm{cape}}$ and $\varepsilon_{l,k}^2$ from (16).
4. Far-field APs: statistical LMMSE [7].

Extra pilot fronthaul: each user forwards $N_{\mathrm{AP}}\,N$ complex scalars to the CPU.
Complexity: $O\big(|\mathcal{X}_k|\,N_{\mathrm{AP}}\,N\big)$ for (13) plus the local OMP, i.e. the same order as polar OMP times a small constant.

6 Data phase: detection (unchanged, cited)

With the estimated channels, each AP forms a local MMSE combiner and the CPU fuses the AP outputs by large-scale fading decoding (LSFD) [1, 2]. For user k served by cluster $\mathcal{S}_k$, with $\hat{\mathbf{H}}_l = [\hat{\mathbf{h}}_{l,1}, \ldots, \hat{\mathbf{h}}_{l,K}]$,

$$\mathbf{v}_{l,k} = \Big(p\,\hat{\mathbf{H}}_l\hat{\mathbf{H}}_l^{\mathsf{H}} + p\sum_j \mathbf{C}_{l,j} + \mathbf{I}_N\Big)^{-1}\hat{\mathbf{h}}_{l,k}, \qquad \hat{x}_k = \sum_{l\in\mathcal{S}_k} a_{l,k}^{\star}\,\mathbf{v}_{l,k}^{\mathsf{H}}\mathbf{y}_l. \tag{20}$$

Hard cases are resolved by the *cross-AP list-based SIC* detector of our prior work [12–14]: a reliability gate fuses all-AP observations only when the local decision is unreliable, giving the fronthaul

$$\bar{B}_k = |\mathcal{S}_k| + \rho\big(L - |\mathcal{S}_k|\big), \tag{21}$$

with ρ the gate-firing rate. CAPE feeds this detector better channels, so ρ (hence the data fronthaul) drops.

7 Why the non-uniform sparse array (ECA) is the right choice

CAPE needs *two* properties at once; only the non-uniform sparse array has both.

(a) Sharp resolution / localization needs a large aperture. Angular resolution scales as $\Delta s \sim \lambda/D$, and near-field *range* focusing is possible only within the beam-depth threshold [3, Prop. 2]

$$r_{l,k} < r_{\mathrm{th}} \approx \frac{D^2\cos^2\theta_{l,k}}{1.6^2\cdot 4}\,\frac{2}{\lambda} = \frac{d_R\cos^2\theta_{l,k}}{10.24}. \tag{22}$$

The position-error floor obeys the near-field Cramér–Rao scaling $\mathrm{std}(\hat{s}) \propto \lambda/(D\sqrt{p})$ and $\mathrm{std}(\hat{r}) \propto r^2/(D^2\sqrt{p})$ [8, 10]: a larger D sharpens *both* angle and range, hence localization. A sparse array buys this large D with the same N (fewer RF chains than a dense array of equal D) [9].

(b) Fusion (19) needs side peaks that differ across APs. A *uniform* sparse array has strong, structured grating lobes (7) (ρ, ρ_m large), so the fusion gain is weak. The *non-uniform* ECA has weak, geometry-dependent side peaks ($\rho_m \ll 1$), so (19) holds. A *dense* array fails (a): short D, poor localization, and $\mathbf{b}_l^{\mathrm{fus}}$ built from a coarse position hurts.

One line. ECA = large aperture (property a) + irregular side peaks (property b); the dense array fails (a), the uniform sparse array fails (b), so CAPE pays off only with the *non-uniform* sparse array.

8 Performance metrics

$$\text{NMSE} = \frac{\sum_{l,k} \left\| \mathbf{h}_{l,k} - \hat{\mathbf{h}}_{l,k} \right\|^2}{\sum_{l,k} \left\| \mathbf{h}_{l,k} \right\|^2}, \qquad \text{SE}_k = \left(1 - \tfrac{\tau}{\tau_c}\right) \log_2\!\left(1 + \gamma_k\right), \tag{23}$$

$$\text{loc. error}_k = \left\| \hat{\mathbf{u}}_k - \mathbf{u}_k \right\|_2, \qquad \bar{B}_k \text{ from (21)}, \quad \text{pilot cost} = N_{\text{AP}} N,$$

with γ_k the post-combining SINR from (20) and τ_c the coherence block length. The information-surplus Δ_k of [13] quantifies the useful energy outside the serving cluster.

Contribution summary. Standard pieces: ECA geometry (1)–(2) [3, 4]; NUSW model (4) [5]; polar OMP (8)–(10) [5, 11]; LSFD/detection (20)–(21) [2, 12, 13]. **New in this work:** the CAPE estimator (12)–(16), the ambiguity- resolution analysis (19), and the array-choice argument of Section 8, i.e. cross-AP position fusion for non-uniform sparse arrays in near-field cell-free XL-MIMO.

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